$\mathcal{K}(\ell_p(\Lambda))$ is relatively injective as a right $\mathcal{B}(\ell_p(\Lambda))$-module

N. T. Nemesh

Abstract: In this note we prove a somewhat unexpected result: the Banach space of compact operators on a reflexive ℓ_p-space is relatively injective as a right Banach module over the algebra of bounded linear operators on the same space.

Keywords: injective Banach module, compact operators, ℓ_p-space.

1 Introduction

Extension problems have been an important topic of functional analysis since its inception. The first example of a successfully solved extension problem is the Hahn-Banach theorem [1, 2, 3]. In modern terms this theorem states that the field of complex numbers is an injective object in the category of Banach spaces. All known injective Banach spaces are isomorphic to the space of continuous functions on some compact space [4]. The property of the object in a given category to be injective depends on the "size" of the category and the "abundance" of morphisms. For example, if one restricts the category of Banach spaces to separable ones, the space c_0 of vanishing sequences becomes injective [5]. Even more, Zippin, showed in [6] that all injective spaces in the category of separable Banach spaces are isomorphic to c_0. Similarly if one restricts the set of morphisms to module maps, without changing the class of objects, then for any set Λ the space $c_0(\Lambda)$ is relatively injective as a right $\ell_\infty(\Lambda)$ module [7]. These results rely on specific Banach geometric properties of c_0-spaces. It is reasonable to expect, that its non-commutative counterpart — the space of compact operators on a Hilbert space $\mathcal{K}(H)$ is not injective, as its geometry is quite distinct. Nevertheless it is true, not only for Hilbert spaces, but for all reflexive ℓ_p-spaces.

2 Notation and definitions

Before we proceed to the main topic we shall give a few definitions.

Let M be a subset of a set N, then χ_M denotes the indicator function of M. If $f : N \to L$ is an arbitrary function, then $f|_M$ denotes its restriction to M.

For a given infinite set Λ we can consider the filter $\mathfrak{F}_\Lambda$ consisting of complements of finite subsets of Λ. It is called the *Frechet filter* on Λ. The standard notion of convergence of sequences is convergence along filter $\mathfrak{F}_\mathbb{N}$. Recall that an ultrafilter $\mathfrak{U}$ on Λ is a free if and only if it contains the Frechet filter $\mathfrak{F}_\Lambda$.

For a given Banach spaces E and F by $\mathcal{B}(E, F)$ we denote the space of all bounded linear operators from E to F. Similarly, $\mathcal{K}(E, F)$ stands for the space of compact operators from E to F. We use shortcuts $\mathcal{B}(E) := \mathcal{B}(E, E)$ and $\mathcal{K}(E) := \mathcal{K}(E, E)$. The symbol 1_E stands for the identity operator on E. By $\langle \cdot, \cdot \rangle : E \times E^* \to \mathbb{C}, (x, f) \mapsto f(x)$ we denote the standard duality between E and E^*. For a given $x \in F$ and $f \in E^*$, by $x \bigcirc f$ we denotes the rank one operator from E to F well defined by equality $(x \bigcirc f)(y) = \langle y, f \rangle x$. One can easily check that $\|x \bigcirc f\| = \|x\|\|f\|$.

For a given familty of Banach spaces $\{E_\lambda : \lambda \in \Lambda\}$ and $p \in [1, +\infty]$ by $\bigoplus_p\{E_\lambda : \lambda \in \Lambda\}$ we denote their ℓ_p-sum. Clearly, $\ell_p(\Lambda) = \bigoplus_p\{\mathbb{C} : \lambda \in \Lambda\}$.

Now we shall remind a few notions from the geometry of Banach spaces. A sequence $(x_n)_{n\in\mathbb{N}}$ in a Banach space E is called *weakly null*, *weak* null* or *norm null* if it converges to zero weakly, weakly* or in norm, respectively. The sequence $(x_n)_{n\in\mathbb{N}}$ is called *semi-normalized* if norms of its elements have positive lower bound and finite upper bound. Let $(\Lambda, <)$ be a well-ordered set. A family of vectors $(e_\lambda)_{\lambda\in\Lambda}$ is called a *Schauder basis* of E if for any vector $x \in E$ there exists a unique family of scalars $(f_\lambda(x))_{\lambda\in\Lambda} \subset \mathbb{C}$ such that

$$x = \sum_{\lambda\in\Lambda} f_\lambda(x) e_\lambda$$

Note that the resulting sum depends on the ordering. If it is not, the basis is called *unconditional*. One can show [[11], theorem 1.1.3] that mappings $(f_\lambda)_{\lambda\in\Lambda}$ are well defined bounded linear functionals. They are called the *biorthogonal functionals* of $(e_\lambda)_{\lambda\in\Lambda}$.

A sequence of vectors $(e_n)_{n\in\mathbb{N}}$ in E is called a *basic sequence* if it is a Schauder basis of its closed linear span. The following construction gives an important source of examples of basic sequences. Suppose $(e_\lambda)_{\lambda\in\Lambda}$ is a Schauder basis of some Banach space E, and let $(\Lambda_n)_{n\in\mathbb{N}}$ is a sequence of disjoint finite subsets of Λ such that $\max(\Lambda_n) < \min(\Lambda_{n+1})$ for all $n \in \mathbb{N}$. The sequence of non-zero vectors $(u_n)_{n\in\mathbb{N}}$ such that u_n lies in the linear span of $(e_\lambda)_{\lambda\in\Lambda_n}$ is called a *block-basic sequence*. Such sequences are indeed basic [[11], lemma 1.3.5]. Suppose E and F — two Banach space, then two sequences $(u_n)_{n\in\mathbb{N}} \subset E$ and $(v_n)_{n\in\mathbb{N}} \subset F$ are called *equivalent* if there is an isomorphism of Banach spaces $T : E \to F$ such that $T(u_n) = v_n$ for all $n \in \mathbb{N}$.

Finally, we give a short introduction into the Banach homology theory. Let A be a Banach algebra. We shall consider only right Banach modules over A with contractive outer action $\cdot : X \times A \to X$. Let X and Y be two right Banach A-modules, then a map $\phi : X \to Y$ is an *A-morphism* if it is a continuous A-module map. Banach A-modules and A-morphisms form a category which we denote by $\mathbf{mod} - A$.

In $\mathbf{mod} - A$ the notion of injectivity can be defined in different ways. We will concern with relative injectivity. Let $\xi : X \to Y$ be an A-morphism. Then it is called *c-relatively admissible* if $T \circ \xi = 1_X$ for some bounded linear operator $T : Y \to X$ with $\|T\| \leq c$. A Banach A-module J is called *C-relatively injective* if for any c-relatively admissible A-morphism $\xi : X \to Y$ and any A-morphism $\phi : X \to J$ there exists a continuous A-module map $\psi : Y \to J$ such that $\|\psi\| \leq cC\|\phi\|$ and the diagram

$$\begin{tikzcd} & Y \arrow[dl, dashed, "\psi"'] \\ J & X \arrow[l, "\phi"] \arrow[u, "\xi"'] \end{tikzcd}$$

commutative. A right Banach A-module is called *relatively injective* if it is C-relatively injective for some $C \geq 1$. As an example, if $A = \{0\}$, the category $\mathbf{mod} - A$ turns into the ordinary category of Banach spaces. In this case all Banach spaces are relatively injective.

3 Main result

In this section we shall prove that for any Hilbert space H the right Banach $\mathcal{B}(H)$-module $\mathcal{K}(H)$ is relatively injective.

Proposition 3.1. *Let Λ be an arbitrary set, then for a any ultrafilter $\mathfrak{U}$ on Λ the map*

$$m_{\mathfrak{U}} : \ell_\infty(\Lambda) \to \mathbb{C}, x \mapsto \lim_{\lambda\to\mathfrak{U}} x(\lambda)$$

is a well defined contractive linear operator of norm one such that $m_{\mathfrak{U}}(\chi_\Lambda) = 1$.

Proof. For any $x \in \ell_\infty(\Lambda)$ the set $\{x(\lambda) : \lambda \in \Lambda\}$ is bounded in $\mathbb{C}$, hence it is a subset of a Hausdorff compact. Therefore [[8], corollary 3.1.11], we have a well defined limit $\lim_{\lambda\to\mathfrak{U}} x(\lambda)$. Hence the map $m_{\mathfrak{U}}$ is well defined. By properties of limits [[8], exercise 3.1.5] the map $m_{\mathfrak{U}}$ is linear, and $m_{\mathfrak{U}}(\chi_\Lambda) = 1$ Continuous functions preserve ultralimits [[8], theorem 3.1.14], so $|m_{\mathfrak{U}}(x)| = \lim_{\lambda\to\mathfrak{U}} |x(\lambda)| \leq \lim_{\lambda\to\mathfrak{U}} \|x\| = \|x\|$. Therfore $m_{\mathfrak{U}}$ is contractive. □

Proposition 3.2. *Let* Λ *be an infinite set, and* $\mathfrak{U}$ *be a free ultrafilter on* Λ*. Then*

(i) $m_{\mathfrak{U}}|_{c_0(\Lambda)} = 0$;

(ii) if $|m_{\mathfrak{U}}(x)| \geq c$ *for some* $x \in \ell_\infty(\Lambda)$ *and* $c > 0$*, then for any* $c' < c$ *the set* $\{\lambda \in \Lambda : |x(\lambda)| > c'\}$ *is infinite.*

Proof. (i) Let $x \in c_0(\Lambda)$, then $\lim_{\lambda\to\mathfrak{F}_\Lambda} x(\lambda) = 0$. Since Λ is infinite, the free ultrafilter $\mathfrak{U}$ contains the Frechet filter $\mathfrak{F}_\Lambda$, so $m_{\mathfrak{U}}(x) = \lim_{\lambda\to\mathfrak{U}} x(\lambda) = \lim_{\lambda\to\mathfrak{F}_\Lambda} x(\lambda) = 0$. Hence $m_{\mathfrak{U}}|_{c_0(\Lambda)} = 0$.

(ii) From the definition of limit along filter the set $A := \{\lambda \in \Lambda : |x(\lambda)| > c'\}$ belongs to $\mathfrak{U}$. Since $\mathfrak{U}$ is a filter, then $\Lambda \setminus A \notin \mathfrak{U}$. A fortiori $\Lambda \setminus A \notin \mathfrak{F}_\Lambda$, so A is infinite. □

Proposition 3.3. *Let* E *be a reflexive Banach space, and* $\mathfrak{U}$ *be an ultrafilter on some set* Λ*. Then there exists a contractive linear operator* $M_{\mathfrak{U}} : \bigoplus_\infty\{E : \lambda \in \Lambda\} \to E$ *such that:*

(i) $\langle M_{\mathfrak{U}}(\xi), f\rangle = m_{\mathfrak{U}}\left(\bigoplus_\infty\{\langle \xi(\lambda), f\rangle : \lambda \in \Lambda\}\right)$ *for all* $\xi \in \bigoplus_\infty\{E : \lambda \in \Lambda\}$ *and* $f \in E^*$;

(ii) $M_{\mathfrak{U}}\left(\bigoplus_\infty\{x : \lambda \in \Lambda\}\right) = x$ *for any* $x \in E$;

Proof. (i) Let $\xi \in \bigoplus_\infty\{E : \lambda \in \Lambda\}$, then $\{\langle \xi(\lambda), f\rangle : \lambda \in \Lambda\} \subset \mathbb{C}$ is bounded, so the map

$$F_{\mathfrak{U},\xi} : E^* \to \mathbb{C},\ f \mapsto m_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty\{\langle \xi(\lambda), f\rangle : \lambda \in \Lambda\}\right)$$

is well defined. Clearly, it is a bounded linear functional with $\|F_{\mathfrak{U},\xi}\| \leq \|\xi\|$. Since E is reflexive, then there exists a unique vector $M_{\mathfrak{U},\xi} \in E$ such that $\|M_{\mathfrak{U},\xi}\| = \|F_{\mathfrak{U},\xi}\|$ and $F_{\mathfrak{U},\xi}(f) = \langle M_{\mathfrak{U},\xi}, f\rangle$ for all $f \in E^*$. Consider map $M_{\mathfrak{U}} : \bigoplus_\infty\{E : \lambda \in \Lambda\} \to E$, $\xi \mapsto M_{\mathfrak{U},\xi}$. It is easy to check that $M_{\mathfrak{U}}$ is a linear operator. This operator is contractive because $\|M_{\mathfrak{U}}(\xi)\| = \|F_{\mathfrak{U},\xi}\| \leq \|\xi\|$. Now we proceed to the proof of the properties of $M_{\mathfrak{U}}$.

(i) This equality directly follows from the properties of the vector $M_{\mathfrak{U},\xi}$.

(ii) From proposition 3.1 for any $x \in E$ and $f \in E^*$ we have

$$\left\langle M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty\{x : \lambda \in \Lambda\}\right), f\right\rangle = m_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty\{\langle x, f\rangle : \lambda \in \Lambda\}\right) = \langle x, f\rangle m_{\mathfrak{U}}(\chi_\Lambda) = \langle x, f\rangle.$$

Since $f \in E^*$ is arbitrary we get the desired equality. □

Proposition 3.4. *Let* E *be a reflexive Banach space. Let* $(e_\lambda)_{\lambda\in\Lambda} \subset E$ *and* $(f_\lambda)_{\lambda\in\Lambda} \subset E^*$ *be two families such that* $\sup\{\|e_\lambda\|\|f_\lambda\| : \lambda \in \Lambda\}$ *is finite. Then for any ultrafilter* $\mathfrak{U}$ *on* Λ *and any bounded linear operator* $\mathcal{T} : \mathcal{B}(E) \to \mathcal{B}(E)$ *the map*

$$T_{\mathfrak{U},\mathcal{T}} : E \to E,\ x \mapsto M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty\{\mathcal{T}(x \bigcirc f_\lambda)(e_\lambda) : \lambda \in \Lambda\}\right)$$

is a well defined bounded linear operator with norm at most $\|\mathcal{T}\| \sup\{\|e_\lambda\|\|f_\lambda\| : \lambda \in \Lambda\}$.

Proof. Let $x \in E$, then denote $\xi_{\mathcal{T},x} = \bigoplus_\infty \{\mathcal{T}(x \bigcirc f_\lambda)(e_\lambda) : \lambda \in \Lambda\}$. Clearly,

$$\sup\{\|\xi_{\mathcal{T},x}(\lambda)\| : \lambda \in \Lambda\} \leq \|\mathcal{T}\|\|x\| \sup\{\|e_\lambda\|\|f_\lambda\| : \lambda \in \Lambda\}$$

so $\xi_{\mathcal{T},x} \in \bigoplus_\infty \{E : \lambda \in \Lambda\}$ and the value $M_{\mathfrak{U}}(\xi_{\mathcal{T},x})$ is well defined. So is the map $T_{\mathfrak{U},\mathcal{T}}$. It is easy to check that $T_{\mathfrak{U},\mathcal{T}}$ is linear. Furthermore, using proposition 3.3 for any $x \in E$ we get

$$\|T_{\mathfrak{U},\mathcal{T}}(x)\| \leq \|M_{\mathfrak{U}}\|\|\xi_{\mathcal{T},x}\| \leq \|\mathcal{T}\|\|x\| \sup\{\|e_\lambda\|\|f_\lambda\| : \lambda \in \Lambda\}.$$

This proves norm estimate for $T_{\mathfrak{U},\mathcal{T}}$. □

Proposition 3.5. *Let E be a reflexive Banach space with a Schauder basis $(e_\lambda)_{\lambda\in\Lambda}$. Let $T : E \to E$ be a non-compact bounded linear operator. Then there exists a weakly null, semi-normalized block-basic sequence $(u_n)_{n\in\mathbb{N}}$ of $(e_\lambda)_{\lambda\in\Lambda}$ such that $(T(u_n))_{n\in\mathbb{N}}$ is semi-normalized.*

Proof. Note that the property of a sequence to be weakly null, norm null or semi-normalized is preserved when passing to subsequences or equivalent sequences. We shall use these facts many times during the proof.

Since T is not compact, then there exists a weakly null sequence $(x_n)_{n\in\mathbb{N}} \subset E$ such that $(T(x_n))_{n\in\mathbb{N}}$ is not norm null. Passing to a subsequence we can assume that $\inf\{\|T(x_n)\| : n \in \mathbb{N}\} > 0$. Note that $(x_n)_{n\in\mathbb{N}}$ cannot be norm null, because otherwise $(T(x_n))_{n\in\mathbb{N}}$ would be norm null. Therefore passing to a subsequence once again we may additionally assume that $\inf\{\|x_n\| : n \in \mathbb{N}\} > 0$. As $(x_n)_{n\in\mathbb{N}}$ is weakly convergent, it is bounded. So is $(T(x_n))_{n\in\mathbb{N}}$, because T is bounded. Thus we have constructed a weakly null, semi-normalized sequence $(x_n)_{n\in\mathbb{N}}$ such that $(T(x_n))_{n\in\mathbb{N}}$ is also semi-normalized.

Since $(x_n)_{n\in\mathbb{N}}$ is a countable family, it lies in a closed linear span of some countable subbasis $(e_\lambda)_{\lambda\in\Lambda'}$ of the basis $(e_\lambda)_{\lambda\in\lambda}$. Now by Bessaga-Pelczyński selection principle [[11] proposition 1.3.10] there exists a block-basic sequence $(u_k)_{k\in\mathbb{N}}$ of $(e_\lambda)_{\lambda\in\Lambda'}$ which is equivalent to some subsequence of $(x_n)_{n\in\mathbb{N}}$. As $(u_k)_{k\in\mathbb{N}}$ is a block-basic sequence of $(e_\lambda)_{\lambda\in\Lambda'}$ it is a block-basic sequence of the superset basis $(e_\lambda)_{\lambda\in\Lambda}$. Recall that $(x_n)_{n\in\mathbb{N}}$ is weakly null, semi-normalized with $(T(x_n))_{n\in\mathbb{N}}$ being also semi-normalized. The sequence $(u_k)_{k\in\mathbb{N}}$ also possesses these properites as it is equivalent to a subsequence of $(x_n)_{n\in\mathbb{N}}$. □

Proposition 3.6. *Let Λ be a set and $p \in (1, +\infty)$, then denote $E = \ell_p(\Lambda)$. Let $(e_\lambda)_{\lambda\in\Lambda} \subset E$ be the standard basis and $(f_\lambda)_{\lambda\in\Lambda} \subset E^*$ be the respective biorthogonal functionals. Then for any (free, if Λ is infinite) ultrafilter $\mathfrak{U}$ on Λ, and any bounded linear operator $\mathcal{T} : \mathcal{B}(E) \to \mathcal{K}(E)$ the map $T_{\mathfrak{U},\mathcal{T}}$ is a compact linear operator of norm at most $\|\mathcal{T}\|$.*

Proof. If E is finite-dimensional, then $\mathcal{B}(E) = \mathcal{K}(E)$, so there is nothing to prove.

Suppose E is infinite dimensional, then Λ is infinite. Denote $T := T_{\mathfrak{U},\mathcal{T}}$ By proposition 3.4 the map T is a bounded linear operator with $\|T\| \leq \|\mathcal{T}\|$. Assume towards a contradiction, that T is not compact. From 3.5 we get a block-basic sequence $(u_n)_{n\in\mathbb{N}}$ of $(e_\lambda)_{\lambda\in\Lambda}$ such that $c := \inf\{\|T(u_n)\| : n \in \mathbb{N}\} > 0$ and $C := \sup\{\|u_n\| : n \in \mathbb{N}\|\} < +\infty$.

Fix $n \in \mathbb{N}$. By Hahn-Banach theorem there exists $g_n \in E^*$ such that $\|g_n\| = 1$ and $\langle T(u_n), g_n\rangle = \|T(u_n)\|$. Consider vector $s_n := \bigoplus_\infty \{\langle \mathcal{T}(u_n \bigcirc f_\lambda)(e_\lambda), g_n\rangle : \lambda \in \Lambda\}$. Note that $s_n \in \ell_\infty(\Lambda)$, because $\sup\{|s_n(\lambda)| : \lambda \in \Lambda\} \leq C\|\mathcal{T}\|$. It is easy to check that $m_{\mathfrak{U}}(s_n) = \langle T(u_n), g_n\rangle$, so $|m_{\mathfrak{U}}(s_n)| = m_{\mathfrak{U}}(s_n) \geq c$. By proposition 3.2 the set $A_n := \{\lambda \in \Lambda : |s_n(\lambda)| > c/2\}$ is infinite. Since sets A_n are infinite for all $n \in \mathbb{N}$, by induction we can construct an infinite sequence of distinct elements $(\lambda_n)_{n\in\mathbb{N}} \subset \Lambda$ such that $\lambda_n \in A_n$ for all $n \in \mathbb{N}$. In other words there exists a sequence of distinct elments $(\lambda_n)_{n\in\mathbb{N}} \subset \Lambda$ such that $\langle \mathcal{T}(u_n \bigcirc f_{\lambda_n})(e_{\lambda_n}), g_n\rangle > c/2$.

Since $(u_n)_{n\in\mathbb{N}}$ is a block basis of the standard basis of E, then supports of these vectors are disjoint, so for any $x \in \ell_\infty(\mathbb{N})$ and $y \in \ell_p(\Lambda)$ we get

$$\sum_{n\in\mathbb{N}} \|x(n)\langle y, f_{\lambda_n}\rangle u_n\| = \left(\sum_{n\in\mathbb{N}} |x(n)|^p |\langle y, f_{\lambda_n}\rangle|^p \|u_n\|^p\right)^{1/p} \leq C\|x\| \left(\sum_{n\in\mathbb{N}} |\langle y, f_{\lambda_n}\rangle|^p\right)^{1/p} \leq C\|x\|\|y\|.$$

This means that the series $\sum_{n\in\mathbb{N}} x(n)\langle y, f_{\lambda_n}\rangle u_n$ is absolutely convergent and a fortiori convergent. Therefore, for each $x \in \ell_\infty(\mathbb{N})$ we have a well define a map:

$$S_x : E \to E,\ y \mapsto \sum_{n\in\mathbb{N}} x(n)(u_n \bigcirc f_{\lambda_n})(y)$$

One can easily check that S_x is a linear operator. From the proof of absolute convergence one can easily derive that $\|S_x\| \leq C\|x\|$. Now for each $n \in \mathbb{N}$ we define a map

$$f_n : \ell_\infty \to \mathbb{C},\ x \mapsto \langle \mathcal{T}(S_x)(e_{\lambda_n}), g_n\rangle$$

Clearly, it is a bounded linear functional. Obviously $(e_{\lambda_n})_{n\in\mathbb{N}}$ is weakly null. By definition of $\mathcal{T}$ the operator $\mathcal{T}(S_x)$ is compact, so the sequence $(\mathcal{T}(S_x)(e_{\lambda_n}))_{n\in\mathbb{N}}$ is norm null. As functionals $(g_n)_{n\in\mathbb{N}} \subset E^*$ are norm bounded, we conclude

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \langle \mathcal{T}(S_x)(e_{\lambda_n}), g_n\rangle = 0.$$

This means that $(f_n)_{n\in\mathbb{N}}$ is a weak* null sequence. On the other hand

$$\sum_{k\in\mathbb{N}} |f_n(\chi_{\{k\}})| \geq |f_n(\chi_{\{n\}})| = |\langle \mathcal{T}(u_n \bigcirc f_{\lambda_n})(e_{\lambda_n}), g_n\rangle| > c/2.$$

So $\liminf_{n\to\infty} \sum_{k\in\mathbb{N}} |f_n(\chi_{\{k\}})| > 0$. This contradicts Phillips' lemma [[9], corollary 3.4]. Therefore $\mathcal{T}$ is compact. ☐

Proposition 3.7. *Let Λ be a set and $p \in (1, +\infty)$, then denote $E = \ell_p(\Lambda)$. Let $(e_\lambda)_{\lambda\in\Lambda} \subset E$ be the standard basis and $(f_\lambda)_{\lambda\in\Lambda} \subset E^*$ be the respective biorthogonal functionals. Let $\mathfrak{U}$ be any (free, if Λ is infinite) ultrafilter on Λ. Define*

$$\tau_{\mathfrak{U}} : \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E)) \to \mathcal{K}(E),\ \mathcal{T} \mapsto T_{\mathfrak{U},\mathcal{T}}.$$

Then $\tau_{\mathfrak{U}}$ is a contractive $\mathcal{B}(E)$-morphism of right $\mathcal{B}(E)$-modules such that $\tau_{\mathfrak{U}} \circ \rho_{\mathcal{K}(E)} = 1_{\mathcal{K}(E)}$, where $\rho_{\mathcal{K}(E)} : \mathcal{K}(E) \to \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E)),\ K \mapsto (T \mapsto K \circ T)$.

Proof. By proposition 3.6 the map $\tau_{\mathfrak{U}}$ is well defined. It is routine to check that $\tau_{\mathfrak{U}}$ is linear. From the same proposition we also get that $\|\tau_{\mathfrak{U}}(\mathcal{T})\| \leq \|\mathcal{T}\|$, so $\tau_{\mathfrak{U}}$ is contractive. For any $\mathcal{T} \in \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E))$, $A \in \mathcal{B}(E)$ and $x \in E$ we have

$$\begin{aligned}
\tau_{\mathfrak{U}}(\mathcal{T} \cdot A)(x) &= M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty \{(\mathcal{T} \cdot A)(x \bigcirc f_\lambda)(e_\lambda) : \lambda \in \Lambda\}\right) \\
&= M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty \{\mathcal{T}(A(x) \bigcirc f_\lambda)(e_\lambda) : \lambda \in \Lambda\}\right) \\
&= \tau_{\mathfrak{U}}(\mathcal{T})(A(x)) \\
&= (\tau_{\mathfrak{U}}(\mathcal{T}) \cdot A)(x).
\end{aligned}$$

Therefore $\tau_{\mathfrak{U}}$ is a right $\mathcal{B}(E)$-module map. Finally, using proposition 3.3 and biorthogonality of the families $(e_\lambda)_{\lambda\in\Lambda}$ and $(f_\lambda)_{\lambda\in\Lambda}$ we get

$$\begin{aligned}\tau_{\mathfrak{U}}(\rho_{\mathcal{K}(H)}(K))(x) &= M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty \{\rho_{\mathcal{K}(H)}(K)(x \bigcirc f_\lambda)(e_\lambda) : \lambda\in\Lambda\}\right)\\ &= M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty \{\langle e_\lambda, f_\lambda\rangle K(x) : \lambda\in\Lambda\}\right)\\ &= M_{\mathfrak{U}}\left(\bigoplus\nolimits_\infty \{K(x) : \lambda\in\Lambda\}\right)\\ &= K(x).\end{aligned}$$

for all $K\in\mathcal{K}(E)$ and $x\in E$. In other words, $\tau_{\mathfrak{U}}\circ\rho_{\mathcal{K}(E)} = 1_{\mathcal{K}(E)}$. □

Theorem 3.8. *Let E be an ℓ_p-space for some $p\in(1,+\infty)$, then the right $\mathcal{B}(E)$-module $\mathcal{K}(E)$ is 1-relatively injective.*

Proof. By assumption $E=\ell_p(\Lambda)$ for some set Λ. Pick any (free, if Λ is infinite) ultrafilter $\mathfrak{U}$ on Λ. By proposition 3.7 there exists a right contractive $\mathcal{B}(E)$-morphism $\tau_{\mathfrak{U}}$ which is a left inverse for $\rho_{\mathcal{K}(E)}:\mathcal{K}(E)\to\mathcal{B}(\mathcal{B}(E),\mathcal{K}(E))$, $K\mapsto(T\mapsto K\circ T)$. By proposition [[10], proposition VII.1.39(III)] the right $\mathcal{B}(E)$-module $\mathcal{K}(E)$ is relatively injective. As $\tau_{\mathfrak{U}}$ is contractive the module $\mathcal{K}(E)$ is 1-relatively injective [[12], proposition 3.8]. □

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Norbert Nemesh, Faculty of Mechanics and Mathematics, Moscow State University, Moscow 119991 Russia

E-mail: nemeshnorbert@yandex.ru